

provide a virtual display to the end user of the technology, without any hassles of maintaining a head mounted display that is entangled with wires.

Furthermore, AR can be extended to various applications in the business domain, specially focused towards the e-commerce solutions. For instance, customers can be provided with virtual trials of the clothes that they wish to try before a purchase in a 3D simulated environment, which would again require services of both AR and VR to populate an environment and for the user to interact with this system.

Venturing into futuristic ideas, a theoretical concept that had been proposed way back in the past but is still far from reality is Programmable Matter. Proposed by American inventor and futurist Raymond Kurzweil in 1999, was the idea of utility fog (also known as foglets).

A utility fog is a hypothetical collection of microbots that can emulate or augment a physical structure. These foglets can be programmed to create virtual environments that are identical to the real ones, which takes us a step closer to the definition of an ideal virtual reality. Foglets are microscopic bots, hence they can replicate any structure as a whole or a part of that structure. Hence, such microbots can be efficiently used in Augmented Reality systems where holograms were used previously. Therefore AR would no longer be a projection, but a physical extension of the existing world which can be programmed according to the user's wish. Conclusively, foglets as a concept are supplementary to both AR and VR systems but as of now they are far from reality and are only a possibility in computer animated movies like Big Hero 6.

In a futuristic world where imagination is reality, an ideal Virtual Reality system would make the real and virtual worlds indistinguishable while an ideal Augmented Reality system would make them inseparable. Even though the aforementioned definition may seem simpler, but we are still miles away from such a theoretically perfect environment of simulation.



A conceptual design of Foglet

AR v/s VR or AR with VR?

A real question that still remains is, whether Augmented Reality and Virtual Reality are two opposing technologies or if these technologies have the potential to complement each and work together harmoniously?